Next-Over Wicket Predictor

Men's IPL 2008-2026 archive. First model development session | October 2026 | Results verified on 2025 only

Project status. Stage 1 data quality and Stage 2 exploratory analyses are complete for development. Stage 3 is now in progress. This session develops Model 2 only: the probability that at least one batting-team wicket occurs in the over immediately following a completed over. The pre-match batter 30+/50+ milestone predictor is still pending.

Checkpoint	Verified status
Historical input	1,243 matches (2008-2026), sourced from the user-provided Cricsheet archive
Eligible wicket model observations	44,161 candidate checkpoints in the full archive, after initial exclusions
Training period	2008-2024 (39,019 checkpoints overall) or 2018-2024 (16,575 checkpoints)
Validation period	2025: 2,561 checkpoints from 73 matches
Final test	2026 RESERVED - not used in training, tuning or scoring
Engineering tests	27 automated tests passed

What makes this a useful academic experiment?

The target is a real observable event, the input features are limited to facts known at the end of the current over, and model evaluation follows a chronological split. We compare model forecasts to a trivial prior wicket frequency, rather than just reporting classifier accuracy.

Scope of this session

Create auditable next-over labels; decide how to handle interrupted and shortened next overs; compare Logistic Regression variants and historical training windows; show probability calibration and by-phase variation; save a development model for a local CLI demo; document assumptions and open weaknesses.

Evidence: Python scripts under src/cricketlens/, pytest unit tests, tracked validation tables in reports/stage_03/session_01/, and diagnostic charts in reports/figures/stage_03/. No deployed web API or final 2027 model is claimed.

What counts as a valid example?

The exact target definition matters more than the number of training rows.

Situation	Handling in Stage 3 Session 01
Completed current over	Eligible for a prediction checkpoint, provided sequence and counts are valid.
Following over has 6 legal deliveries	Known Yes or No: did a team wicket occur in that over?
Following over ends early AND a wicket occurred	Known Yes: the target event has already happened.
Following over ends early with NO wicket	Censored: exclude, because the full six-ball outcome is unobserved.
Match outcome marked D/L	Exclude from this first ML cohort; preserve in EDA and archive.
Super Over tie-break	Exclude Super Over deliveries, keep eligible ordinary innings.
Suspicious 7-legal-ball recorded over	Exclude corrupted/gapped checkpoints and their contaminated successors.

Audit counts from full archive

45,268 completed checkpoints considered; 619 D/L-marked checkpoint exclusions; 14 anomalous/gapped exclusions; 474 censored incomplete next overs with no wicket; 237 known positives from shortened next overs; 43,924 full-next-over examples. This yields 44,161 eligible records, of which 41,580 are before 2026.

Important caution

The eligible cohort is not a random sample of all imaginable IPL innings situations. Rain truncation, chases ending early, and the decision to exclude incomplete negative windows can change event prevalence and model generalizability. There is no claim to predict the unobserved balls that were never played.

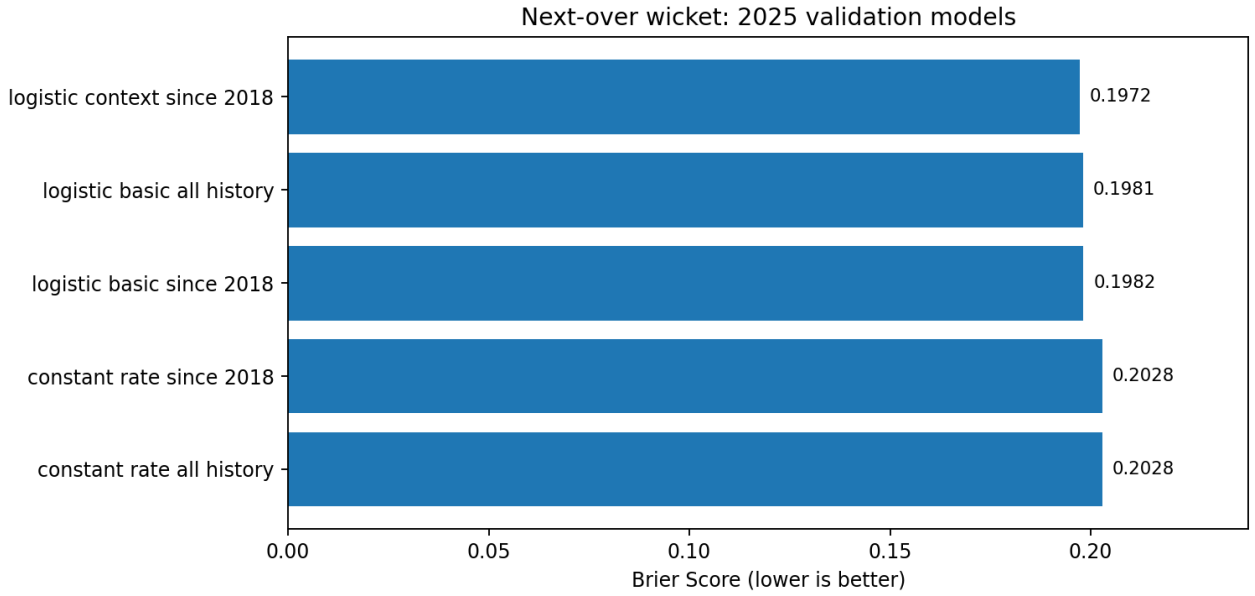
The label_audit fields next_over_legal_balls and label_evidence appear in the exported table only to verify target construction; neither enters the prediction features.

Results and model selection

Three Logistic Regression candidates and two constant-rate references were compared on the same 2,561 examples.

Candidate	Brier ↓	Log loss ↓	ROC-AUC
Context model, 2018-2024	0.197195	0.582958	0.5961
Basic model, 2008-2024	0.198138	0.585367	0.5906
Basic model, 2018-2024	0.198169	0.585416	0.5904
Constant wicket rate, 2018-2024	0.202796	0.595524	0.5000
Constant wicket rate, 2008-2024	0.202811	0.595561	0.5000

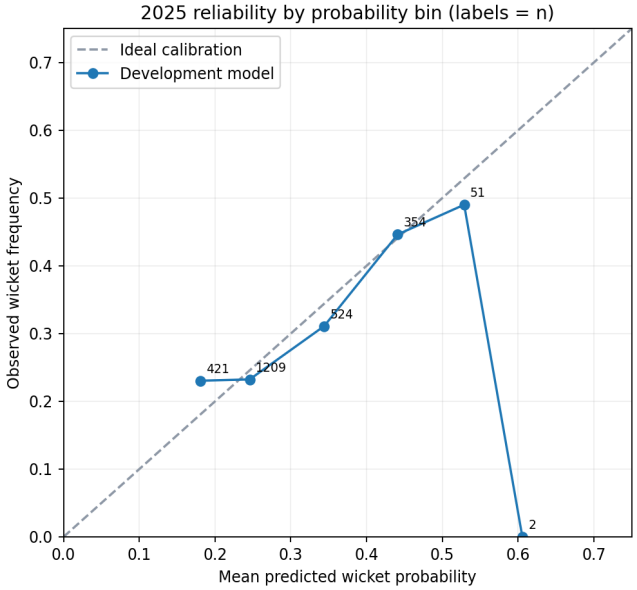
Selected development candidate: contextual Logistic Regression using 2018-2024 training data. Its 2025 Brier improvement against a recent-history constant-rate baseline is 0.005601. A match-cluster bootstrap produced a descriptive 95% interval of [0.002358, 0.008731], but the same 2025 season was used to choose the model, so this is not independent test evidence.

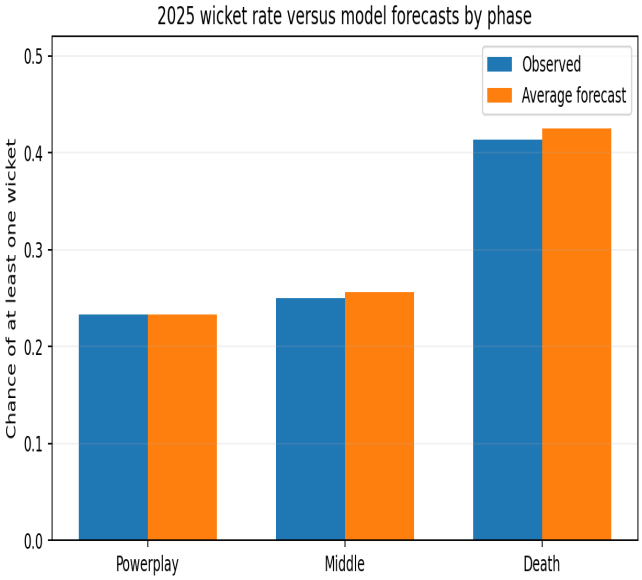


Brier Score measures average squared probability error. Lower values are better. An 0.197 score is NOT 80.3% accuracy. The measured improvement is useful but modest, and high predictive reliability remains unproven.

Calibration and deployment readiness

Does a predicted risk level match the observed wicket frequency over many examples?





On the 2025 examples, wicket event frequency is 23.36% in upcoming Powerplay overs (n=715), 25.04% in middle overs (n=1,266) and 41.38% in death overs (n=580). The model predicts approximately 23.35%, 25.60% and 42.54%, respectively. Sparse high-risk bins require caution.

Implementation and usability

The selected model uses innings, completed overs, runs, wickets, run rate, last-over runs and wickets, batting team, bowling team, phase and venue. The eventual manual-input website will need the last-over fields too. The local development-only demo was smoke-tested for a hypothetical scenario; it is not a live-match forecast.

Next milestones

Session 02: train the pre-match batter 30+/50+ milestone model using only form available before each match, handle playing-XI/DNB, and compare calibrated baselines. Session 03: freeze both models, evaluate 2026 once, document generalization and model limitations, then separately refit through 2026 to support future forecasts. Continue GitHub updates on a feature branch until reviewed and merged.

Academic integrity: Cricsheet must be credited; code, observations and limitations should be understood and verified by the project team. AI use should be disclosed under course guidelines.